Joint convexity of the Umegaki relative entropy

The Lean proof of `qRelativeEnt_joint_convexity`, written conventionally

This note renders the Lean proof of `qRelativeEnt_joint_convexity` (`QuantumInfo/Entropy/DPI.lean`, line ~1534 as of commit `720c9fff`) as a mathematician would write it on paper. The structure of the argument below follows the Lean proof step for step; Lean identifiers are given in `code font` at the point where they are used. Nothing is included that is not in the formal proof, and the two pre-existing ingredients it relies on (joint convexity of the trace functional, and continuity in α) are stated as propositions with their own provenance.

1. Setting and conventions

Throughout, d is a finite index set and states live on $\mathbb{C}^d$.

- A **state** (`MState d`, `QuantumInfo/States/Mixed/MState.lean:63`) is a Hermitian matrix $\rho \geq 0$ with $\operatorname{Tr}\rho = 1$.
- For $A \geq 0$ and $r \in \mathbb{R}$, the power A^r is defined by functional calculus with $0^r := 0$ for $r \neq 0$ (`HermitianMat.rpow`, `QuantumInfo/ForMathlib/HermitianMat/Rpow.lean:45`); negative powers are therefore Moore–Penrose-style powers on the support. Likewise $\log A$ is taken on the support.
- We write $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ for the support (= range) inclusion of PSD matrices; in Lean this is the kernel inclusion σ`.M.ker` $\leq$ ρ`.M.ker`.
- Entropies take values in $[0, \infty]$ (ℝ≥0∞), with the conventions $a \cdot \infty = \infty$ for $a \neq 0$ and $0 \cdot \infty = 0$.
- p ranges over `Prob` $= \{p \in \mathbb{R} \mid 0 \leq p \leq 1\}$ (`QuantumInfo/ClassicalInfo/Prob.lean:36`). The mixture notation `p` $[\rho_1 \leftrightarrow \rho_2]$ (`Mixable.mix`, `Prob.lean:267`) denotes the state whose matrix is $p\rho_1 + (1-p)\rho_2$. In the theorem statement, `1 - p` on the right-hand side is truncated subtraction in ℝ≥0∞, which equals the real $1 - p$ since $p \leq 1$.
- log is the natural logarithm.

2. Definitions

Definition 1 (Umegaki relative entropy; `qRelativeEnt`, notation D, `Relative.lean:1497`).

$$D(\rho \,\|\, \sigma) = \begin{cases} \operatorname{Tr}\big[\rho\,(\log\rho - \log\sigma)\big] & \text{if } \operatorname{supp}\rho \subseteq \operatorname{supp}\sigma, \\ +\infty & \text{otherwise.} \end{cases}$$

(In Lean, $\mathbf{D}$`(`$\rho\|\sigma$`)` is *defined* as `D_1(`$\rho\|\sigma$`)`, the $\alpha = 1$ member of the next definition; the identification with the trace formula under the support condition is `qRelativeEnt_ker`, `Relative.lean:1782`.)

Definition 2 (sandwiched Rényi relative entropy; `SandwichedRelRentropy`, notation `D_`α, `Relative.lean:1480`). For $\alpha > 0$, set $\gamma := \frac{1-\alpha}{2\alpha}$. If $\operatorname{supp}\rho \not\subseteq \operatorname{supp}\sigma$, then $\widetilde{D}_\alpha(\rho\|\sigma) := +\infty$. Otherwise

$$\widetilde{D}_\alpha(\rho \,\|\, \sigma) \;:=\; \begin{cases} \dfrac{1}{\alpha-1}\log \operatorname{Tr}\big[(\sigma^\gamma \rho\, \sigma^\gamma)^\alpha\big] & \alpha \neq 1, \\ \operatorname{Tr}\big[\rho\,(\log\rho - \log\sigma)\big] & \alpha = 1, \end{cases}$$

a nonnegative quantity (`sandwichedRelRentropy_nonneg`). (For $\alpha \le 0$ the Lean definition returns the junk value 0; only $\alpha \ge 1$ is used here.)

Definition 3 (sandwiched trace functional; `sandwichedTraceFunctional`, notation `Q_`α, `DPI.lean:66`). For $\alpha \in \mathbb{R}$ and $\gamma = \frac{1-\alpha}{2\alpha}$,

$$\widetilde{Q}_\alpha(\rho \,\|\, \sigma) \;:=\; \operatorname{Tr}\big[(\sigma^\gamma \rho\, \sigma^\gamma)^\alpha\big] \;\in\; \mathbb{R}.$$

For $0 < \alpha \neq 1$ and $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ one has (`sandwichedRelRentropy_eq_log_traceFunctional`, `DPI.lean:78`)

$$\widetilde{D}_\alpha(\rho\|\sigma) = \Big[\tfrac{1}{\alpha-1}\log \widetilde{Q}_\alpha(\rho\|\sigma)\Big]^+ \quad \text{(as an element of } [0,\infty],\text{ via ofReal)}.$$

3. Theorem

Theorem (`qRelativeEnt_joint_convexity`, `DPI.lean:1522`). For all states $\rho_1, \rho_2, \sigma_1, \sigma_2$ on $\mathbb{C}^d$ and all $p \in [0,1]$, writing $\bar\rho := p\rho_1 + (1-p)\rho_2$ and $\bar\sigma := p\sigma_1 + (1-p)\sigma_2$,

$$D(\bar\rho \,\|\, \bar\sigma) \;\le\; p\, D(\rho_1\|\sigma_1) \;+\; (1-p)\, D(\rho_2\|\sigma_2)$$

as an inequality in $[0,\infty]$.

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theorem qRelativeEnt_joint_convexity :
    ∀ (ρ₁ ρ₂ σ₁ σ₂ : MState d), ∀ (p : Prob),
      𝐃(p [ρ₁ ↔ ρ₂]‖p [σ₁ ↔ σ₂]) ≤ p * 𝐃(ρ₁‖σ₁) + (1 - p) * 𝐃(ρ₂‖σ₂)
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4. Ingredients

The proof consumes two substantial pre-existing results (A, B), one elementary matrix lemma (C), and two scalar inequalities from Mathlib (D). Only Lemma E and the assembly in §5 are new in this contribution.

Proposition A (joint convexity of $\widetilde{Q}_\alpha$, $\alpha > 1$; `sandwichedTraceFunctional_jointly_convex`, `DPI.lean:568`, pre-existing). Let $\alpha > 1$, let ι be a finite index set, let $w_i \ge 0$ with $\sum_i w_i = 1$, and let (ρ_i), (σ_i) be states with $\operatorname{supp}\rho_i \subseteq \operatorname{supp}\sigma_i$ for every i. If $\bar\rho, \bar\sigma$ are states with $\bar\rho = \sum_i w_i\rho_i$ and $\bar\sigma = \sum_i w_i\sigma_i$ as matrices, then

$$\widetilde{Q}_\alpha(\bar\rho \,\|\, \bar\sigma) \;\le\; \sum_i w_i\, \widetilde{Q}_\alpha(\rho_i\|\sigma_i).$$

Proof route (already formalized; sketch only). For $\alpha > 1$ and $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ one has the variational formula (`traceFunctional_eq_iSup_f_alpha`, `DPI.lean:434`)

$$\widetilde{Q}_\alpha(\rho\|\sigma) = \sup_{H \ge 0} f_\alpha(H,\rho,\sigma), \qquad f_\alpha(H,\rho,\sigma) = \alpha \operatorname{Tr}[\rho H] - (\alpha-1)\operatorname{Tr}\big[(\sigma^{-\gamma} H \sigma^{-\gamma})^{\alpha/(\alpha-1)}\big]$$

(`f_alpha`, `DPI.lean:174`), the supremum being attained at $\hat{H} = \sigma^{\gamma}(\sigma^{\gamma}\rho\,\sigma^{\gamma})^{\alpha-1}\sigma^{\gamma}$. For fixed $H \geq 0$, f_α is linear in ρ and convex in σ: with $s := -\gamma = \frac{\alpha-1}{2\alpha} \in (0, \frac{1}{2})$ and $q := \frac{\alpha}{\alpha-1} > 1$, the map $\sigma \mapsto \mathrm{Tr}[(\sigma^s H \sigma^s)^q]$ is concave on the PSD cone (`HermitianMat.trace_conj_rpow_concave`, `QuantumInfo/ForMathlib/HermitianMat/LiebConcavity.lean:494`, itself derived from the Lieb–Ando trace theorems formalized in `QuantumInfo/ForMathlib/HayataGroup/TraceInequality/LiebAndoTrace.l` and it enters f_α with the negative coefficient $-(\alpha-1)$. Hence each $f_\alpha(H,\cdot,\cdot)$ is jointly convex (`f_alpha_jointly_convex`, `DPI.lean:494`), and a pointwise supremum of jointly convex functions is jointly convex (`iSup_f_alpha_jointly_convex`, `DPI.lean:527`). This is the Frank–Lieb variational argument in the form used by Leditzky–Rouzé–Datta; see the literature note. ☐

Proposition B (continuity in α; `sandwichedRelRentropy.continuousOn`, `Relative.lean:1769`, pre-existing). For fixed states ρ, σ, the map $\alpha \mapsto \widetilde{D}_\alpha(\rho\|\sigma)$ is continuous on $(0,\infty)$ (with respect to the order topology on $[0,\infty]$). The consequence used below is:

$$\widetilde{D}_\alpha(\rho\|\sigma) \longrightarrow \widetilde{D}_1(\rho\|\sigma) = D(\rho\|\sigma) \qquad (\alpha \to 1^+).$$

Note that since $\widetilde{D}_1 := D$ *by definition*, the $\alpha \to 1$ convergence of the sandwiched Rényi entropy to the Umegaki entropy is precisely the content of this continuity theorem; it is a genuine analytic fact proved earlier in the library, not a definitional triviality.

Lemma C (kernels of weighted sums; `HermitianMat.ker_weighted_sum_le`, `DPI.lean:545`). If $w_i \geq 0$, all $\rho_i, \sigma_i \geq 0$, and $\operatorname{supp}\rho_i \subseteq \operatorname{supp}\sigma_i$ for all i, then $\operatorname{supp}(\sum_i w_i\rho_i) \subseteq \operatorname{supp}(\sum_i w_i\sigma_i)$.

Scalar facts D (Mathlib). (i) $\log x \leq x - 1$ for $x > 0$ (`Real.log_le_sub_one_of_pos`). (ii) $|e^x - 1 - x| \leq x^2$ for $|x| \leq 1$ (`Real.abs_exp_sub_one_sub_id_le`); in particular $e^x - 1 \leq x + x^2$ there.

Also used: strict positivity $\widetilde{Q}_\alpha(\rho\|\sigma) > 0$ under the support condition (`sandwichedTraceFunctional_pos`, `DPI.lean:100`).

5. The key new lemma: a convergent majorant for the difference quotient

Lemma E (`sandwichedTraceFunctional_sub_one_div_eventually_le`, `DPI.lean:1432`, new). Let ρ, σ be states with $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ (so $D(\rho\|\sigma) < \infty$). Then there is a function $u : \mathbb{R} \to \mathbb{R}$ with

$$u(\alpha) \xrightarrow[\alpha \to 1^+]{} D(\rho\|\sigma) \qquad \text{and} \qquad \frac{\widetilde{Q}_\alpha(\rho\|\sigma) - 1}{\alpha - 1} \leq u(\alpha) \quad \text{for all } \alpha > 1 \text{ sufficiently close to 1.}$$

Proof. For $\alpha > 1$ put

$$r(\alpha) := \frac{\log \widetilde{Q}_\alpha(\rho\|\sigma)}{\alpha - 1}, \qquad x(\alpha) := (\alpha - 1)\,r(\alpha) = \log \widetilde{Q}_\alpha(\rho\|\sigma),$$

and define $u(\alpha) := r(\alpha) + x(\alpha)\,r(\alpha) = r(\alpha) + (\alpha - 1)r(\alpha)^2$.

1. Under the support condition, $r(\alpha) \geq 0$ and $\widetilde{D}_\alpha(\rho\|\sigma) = [r(\alpha)]^+$ (Definition 3 with `sandwichedRelRentropy_nonneg`), and $D(\rho\|\sigma) \neq \infty$ (Lean: `qRelativeEnt_ne_top_iff`, `Relative.lean:1789`). By Proposition B — packaged as the named lemma `sandwichedRelRentropy_tendst` (`DPI.lean:1423`), which states $\widetilde{D}_\alpha(\rho\|\sigma) \to D(\rho\|\sigma)$ as $\alpha \to 1^+$ — we get $r(\alpha) \to D(\rho\|\sigma)$ (as real numbers) (Lean: `h_r_tendsto`, passing through `ENNReal.tendsto_toReal`).
2. Hence $x(\alpha) = (\alpha - 1)r(\alpha) \to 0$, so eventually $|x(\alpha)| \leq 1$ (Lean: `h_eps`, `h_small`).

3. $\widetilde{Q}_\alpha(\rho\|\sigma) > 0$ (positivity above), so $\widetilde{Q}_\alpha(\rho\|\sigma) = e^{x(\alpha)}$. By D(ii), for the eventual α of step 2, $\widetilde{Q}_\alpha - 1 \le x(\alpha) + x(\alpha)^2$; dividing by $\alpha - 1 > 0$,

$$\frac{\widetilde{Q}_\alpha(\rho\|\sigma) - 1}{\alpha - 1} \;\le\; r(\alpha) + x(\alpha)\, r(\alpha) \;=\; u(\alpha).$$

4. By steps 1–2, $u(\alpha) \to D(\rho\|\sigma) + 0 \cdot D(\rho\|\sigma) = D(\rho\|\sigma)$. ■

Heuristic remark (not part of the formal proof). Since $\widetilde{Q}_1(\rho\|\sigma) = \operatorname{Tr}\rho = 1$, the quantity $(\widetilde{Q}_\alpha - 1)/(\alpha - 1)$ is a difference quotient whose two-sided limit "should" be $\frac{d}{d\alpha}\widetilde{Q}_\alpha\big|_{\alpha=1} = D(\rho\|\sigma)$. The formal proof deliberately avoids establishing differentiability: a one-sided *eventual upper bound by a convergent majorant* suffices for the limit argument, and that follows from continuity of $\widetilde{D}_\alpha$ alone, via the two elementary scalar bounds.

6. Proof of the theorem

Fix $\rho_1, \rho_2, \sigma_1, \sigma_2$ and p. The proof splits exactly as the Lean source does.

Step 0 — degenerate weights. If $p = 0$, then $\bar\rho = \rho_2$, $\bar\sigma = \sigma_2$ (in Lean via `Mixable.mix_zero`, `Prob.lean:279`), and the right-hand side is $0 \cdot D(\rho_1\|\sigma_1) + 1 \cdot D(\rho_2\|\sigma_2) = D(\rho_2\|\sigma_2)$ — note this uses $0 \cdot \infty = 0$, so the case is genuine, not vacuous. Equality holds. If $p = 1$, symmetrically $\bar\rho = \rho_1$, $\bar\sigma = \sigma_1$ via the mixture identity `1 [x₁ ↔ x₂] = x₁`, now the `@[simp]` lemma `Mixable.mix_one` (`Prob.lean:284`), and the right side is $D(\rho_1\|\sigma_1)$. In Lean both degenerate cases now close by a single `simp` (the `mix_zero` / `mix_one` simp lemmas discharge the mixture, and the $0 \cdot \infty = 0$ / $1 \cdot x = x$ arithmetic follows). Henceforth $0 < p < 1$ (Lean: `hp0'`, `hp1'`).

Step 1 — infinite right-hand side. Suppose $\operatorname{supp}\rho_1 \not\subseteq \operatorname{supp}\sigma_1$ (Lean: `hker₁` fails). Then $D(\rho_1\|\sigma_1) = \infty$ by Definition 1 — in Lean via the ⊤-characterization `qRelativeEnt_eq_top_iff` (`Relative.lean:1795`) — and since $p \ne 0$, $p \cdot \infty = \infty$, so the right-hand side is ∞ and the inequality is trivial (`le_top`). The case $\operatorname{supp}\rho_2 \not\subseteq \operatorname{supp}\sigma_2$ is identical, using $1 - p \ne 0$. So we may assume both support conditions

$$\operatorname{supp}\rho_i \subseteq \operatorname{supp}\sigma_i \quad (i = 1, 2),$$

whence $D(\rho_1\|\sigma_1), D(\rho_2\|\sigma_2) < \infty$; in Lean this finiteness is supplied at point of use by the companion lemma `qRelativeEnt_ne_top_iff` (`Relative.lean:1789`) rather than a standing `have`.

Step 2 — setup for the main case. Write the mixtures as weighted sums over $\iota = \{1, 2\}$ (Lean: `Fin 2`) with weight vector $w = (p,\, 1 - p)$:

$$\bar\rho = \textstyle\sum_i w_i \rho_i, \qquad \bar\sigma = \sum_i w_i \sigma_i$$

as matrix identities. This unfolding of `Mixable.mix` into a `Fin 2` weighted sum, with $w_i \ge 0$ and $\sum_i w_i = 1$, is now the standalone lemma `mix_M_eq_weighted_sum` (`DPI.lean:1485`), replacing what were the inline `haves` `hM_ρ`, `hM_σ` (and the side facts `hw_nonneg`, `hw_sum`, now discharged locally inside that lemma and its consumers). By Lemma C, $\operatorname{supp}\bar\rho \subseteq \operatorname{supp}\bar\sigma$; this support-preservation of a binary mixture is likewise extracted as the lemma `ker_mix_le` (`DPI.lean:1493`), which absorbs the former kernel bookkeeping (`hkers`), so the theorem now simply sets `hker_mix := ker_mix_le p hker₁ hker₂` (Lean: `hker_mix`). Apply Lemma E to each pair to obtain majorants u_1, u_2 with $u_i(\alpha) \to D(\rho_i\|\sigma_i)$ as $\alpha \to 1^+$ and, eventually, $(\widetilde{Q}_\alpha(\rho_i\|\sigma_i) - 1)/(\alpha - 1) \le u_i(\alpha)$.

Step 3 — pointwise bound for $\alpha > 1$ (Lean: `h_ev`). For $\alpha > 1$ in the eventual range of both majorants, using $\widetilde{Q}_\alpha(\bar\rho\|\bar\sigma) > 0$ (positivity, with `hker_mix`) and Definition 3:

$$\widetilde{D}_\alpha(\bar\rho\|\bar\sigma) = \Big[\frac{\log \widetilde{Q}_\alpha(\bar\rho\|\bar\sigma)}{\alpha - 1}\Big]^+,$$

and we bound the bracketed real number:

$$\begin{aligned}
\frac{\log \widetilde{Q}_\alpha(\bar\rho\|\bar\sigma)}{\alpha-1} &\le \frac{\widetilde{Q}_\alpha(\bar\rho\|\bar\sigma) - 1}{\alpha - 1} && \text{[D(i): } \log x \le x - 1\text{]}\\
&\le \frac{p\,\widetilde{Q}_\alpha(\rho_1\|\sigma_1) + (1-p)\,\widetilde{Q}_\alpha(\rho_2\|\sigma_2) - 1}{\alpha - 1} && \text{[Prop. A, binary form]}\\
&= p\,\frac{\widetilde{Q}_\alpha(\rho_1\|\sigma_1) - 1}{\alpha - 1} + (1-p)\,\frac{\widetilde{Q}_\alpha(\rho_2\|\sigma_2) - 1}{\alpha - 1} && \text{[since } p + (1-p) = 1\text{]}\\
&\le p\,u_1(\alpha) + (1-p)\,u_2(\alpha) && \text{[Lemma E].}
\end{aligned}$$

Hence, eventually as $\alpha \to 1^+$,

$$\widetilde{D}_\alpha(\bar\rho\|\bar\sigma) \;\le\; \big[\,p\,u_1(\alpha) + (1-p)\,u_2(\alpha)\,\big]^+.$$

The binary-mixture instance of Proposition A used in the second inequality is now the named lemma `sandwichedTraceFunctional_mix_le` (`DPI.lean:1504`) — the former inline `have h_convex` — which specializes `sandwichedTraceFunctional_jointly_convex` to a `Mixable` mixture over `Fin 2` (via `mix_M_eq_weighted_sum`). Positivity uses `hker_mix`; the whole chain is the Lean `have h_ev`.

Step 4 — pass to the limit $\alpha \to 1^+$. By Proposition B (applied to the mixture pair) — invoked as `sandwichedRelRentropy_tendsto_qRelativeEnt` (`DPI.lean:1423`) — the left side tends to $D(\bar\rho\|\bar\sigma)$ (Lean: `h_lhs`). By Lemma E and continuity of $t \mapsto [t]^+$, the right side tends to

$$\big[\,p\,D(\rho_1\|\sigma_1) + (1-p)\,D(\rho_2\|\sigma_2)\,\big]^+ \;=\; p\,D(\rho_1\|\sigma_1) + (1-p)\,D(\rho_2\|\sigma_2)$$

(Lean: `h_rhs`). The identification of the $[\,\cdot\,]^+$ of the real convex combination with the $[0,\infty]$-valued convex combination is the inline `have h_id` (`DPI.lean:1561`), whose hypotheses are exactly the two finiteness facts from Step 1 (supplied by `qRelativeEnt_ne_top_iff`). Since the inequality of Step 3 holds eventually along the (nontrivial) filter of right-neighbourhoods of 1, the inequality passes to the limits (`le_of_tendsto_of_tendsto`):

$$D(\bar\rho\|\bar\sigma) \;\le\; p\,D(\rho_1\|\sigma_1) + (1-p)\,D(\rho_2\|\sigma_2). \qquad \blacksquare$$

7. Lean–paper dictionary

Paper object / step	Lean identifier	Location (commit `720c9fff`)
Umegaki D	`qRelativeEnt` / **D**	`Relative.lean:1497`
— trace-formula identification	`qRelativeEnt_ker`	`Relative.lean:1782`
Sandwiched $\widetilde{D}_\alpha$	`SandwichedRelRentropy` / `D_α`	`Relative.lean:1480`
Trace functional $\widetilde{Q}_\alpha$	`sandwichedTraceFunctional` / `Q_α`	`DPI.lean:66`

Paper object / step	Lean identifier	Location (commit `720c9fff`)
$\widetilde{D}_\alpha = [\log \widetilde{Q}_\alpha/(\alpha{-}1)]^+$	`sandwichedRelRentropy_eq_log_traceFunctional`	`DPI.lean:78`
Mixture $p[\cdot \leftrightarrow \cdot]$	`Mixable.mix`	`Prob.lean:267`
— degenerate $p = 0$ / $p = 1$	`Mixable.mix_zero` / `Mixable.mix_one` (NEW `@[simp]`)	`Prob.lean:279, 284`
Proposition A	`sandwichedTraceFunctional_jointly_convex`	`DPI.lean:568`
— variational formula	`traceFunctional_eq_iSup_f_alpha`, `f_alpha`, `H_hat`	`DPI.lean:434, 174, 180`
— concavity in σ	`HermitianMat.trace_conj_rpow_concave`	`LiebConcavity.lean:494`
Proposition B (continuity)	`sandwichedRelRentropy.continuousOn`	`Relative.lean:1769`
— as $\alpha \to 1^+$ limit (NEW)	`sandwichedRelRentropy_tendsto_qRelativeEnt`	`DPI.lean:1423`
Lemma C	`HermitianMat.ker_weighted_sum_le`	`DPI.lean:545`
D(i)	`Real.log_le_sub_one_of_pos`	Mathlib
D(ii)	`Real.abs_exp_sub_one_sub_id_le`	Mathlib
$\widetilde{Q}_\alpha > 0$	`sandwichedTraceFunctional_pos`	`DPI.lean:100`
Lemma E	`sandwichedTraceFunctional_sub_one_div_eventually_le`	`DPI.lean:1432`
Finiteness $D \neq \infty$ / $D = \infty$ (NEW)	`qRelativeEnt_ne_top_iff` / `qRelativeEnt_eq_top_iff`	`Relative.lean:1789, 1795`
Step 2, mixture as `Fin 2` sum (NEW; was `hM_`ρ,`hM_`σ)	`mix_M_eq_weighted_sum`	`DPI.lean:1485`
Step 2, support of mixture (NEW; was `hkers`)	`ker_mix_le`	`DPI.lean:1493`
Step 3, binary Prop. A (NEW; was `h_convex`)	`sandwichedTraceFunctional_mix_le`	`DPI.lean:1504`
Step 4, ofReal convex-combo	inline `have h_id`	`DPI.lean:1561`
Steps 0–4	`qRelativeEnt_joint_convexity`	`DPI.lean:1522`
Step 3	`h_ev` (calls `sandwichedTraceFunctional_mix_le`)	in theorem, `DPI.lean:1522+`
Step 4	`h_lhs`, `h_rhs`, `le_of_tendsto_of_tendsto`	in theorem, `DPI.lean:1522+`

8. Remarks for the reviewer

- **What is new.** Lemma E and the theorem itself (§5–§6), together with the supporting lemmas it is factored into (the public `sandwichedRelRentropy_tendsto_qRelativeEnt` and the private `mix_M_eq_weighted_sum`, `ker_mix_le`, `sandwichedTraceFunctional_mix_le`), account for +190 lines in `DPI.lean`. The previously `sorry`-stubbed statement was removed from `Relative.lean` (+11 / −15), where two small finiteness lemmas `qRelativeEnt_ne_top_iff` / `qRelativeEnt_eq_top_iff` were added; and a one-line `@[simp]` lemma `Mixable.mix_one` (+5) was added to `Prob.lean` — net +206 / −15 across the three files. Propositions A and B and all deeper inputs (Lieb–Ando trace theorems) were already in the library. The theorem was placed in `DPI.lean` rather than `Relative.lean` because the $\widetilde{Q}_\alpha$ machinery lives there and importing it from `Relative.lean` would create a cycle.
- **No differentiability is needed.** The classical way to see $\widetilde{D}_\alpha \to D$ and to transfer convexity is to differentiate $\alpha \mapsto \widetilde{Q}_\alpha$ at $\alpha = 1$; the formal proof replaces this by the one-sided majorant

of Lemma E, which needs only continuity (Proposition B) plus the scalar bounds D. This is the main proof-engineering choice.

- **The statement is the binary form.** Finite mixtures of n states follow by induction in the usual way, but only the n-ary form of Proposition A (arbitrary finite `Fintype ι`) is formalized; the **D**-level statement is for two pairs, matching the pre-existing `sorryful` statement it closes.
- **Axioms.** The proof is `sorry`-free; `#print axioms qRelativeEnt_joint_convexity` reports only `propext`, `Classical.choice`, `Quot.sound` (recorded in the commit message of `720c9fff`).